



## The 66th ASH Annual Meeting Abstracts

## POSTER ABSTRACTS

## 701. EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

**Local Myeloid Network Resilience Determines Organ-Specific Acute GvHD Outcomes**

Michael AG Kern<sup>1</sup>, R  mi Doucet Ladev  ze<sup>2</sup>, Haroon Shaikh, PhD<sup>1</sup>, Mariachiara Aloisi<sup>3</sup>, Muthita Khamwong<sup>3</sup>, Juan Vargas<sup>4</sup>, Shreeya Tavkari<sup>3</sup>, Hermann Einsele, MD<sup>5</sup>, Mercedes Gomez de Ag  ero, PhD<sup>3</sup>, Andreas Beilhack<sup>6</sup>

<sup>1</sup>Department of Medicine II, University Hospital W  rzburg, W  rzburg, Germany

<sup>2</sup>Institute for Systemsimmunology, University of W  rzburg, W  rzburg, Germany

<sup>3</sup>University W  rzburg, W  rzburg, Germany

<sup>4</sup>University Hospital W  rzburg, W  rzburg, Germany

<sup>5</sup>Department of Internal Medicine II, University Hospital Wuerzburg, Wuerzburg, Germany

<sup>6</sup>Department of Internal Medicine II, University Hospital W  rzburg, W  rzburg, Germany

Acute graft-versus-host disease (aGvHD) remains a major complication in patients undergoing allogeneic hematopoietic cell transplantation (allo-HCT). Notably, aGvHD predominantly damages the skin, liver, and gastrointestinal tract while sparing other host organ systems. The mechanisms behind this remarkable organotropism in aGvHD have largely remained elusive. Recently, we discovered that disruption of PD-L1<sup>hi</sup> tissue-resident intestinal macrophage networks exacerbates aGvHD. To elucidate the role of myeloid networks in organotropic aGvHD pathogenesis, we investigated the myeloid compartments within aGvHD target and non-target organs after allo-HCT, considering microbial colonization and antibiotic disruption of the microbiota.

To reveal alterations in the myeloid compartments of aGvHD target organs (intestines, skin and liver) as well as non-target organs such as the kidney, we employed light sheet fluorescence microscopy, confocal microscopy, spectral flow cytometry and single-cell RNA sequencing in mouse models of allo-HCT (FVB/N, H2<sup>a</sup>  C57Bl/6, H2<sup>b</sup>, 9 Gy TBI) to uncover changes in the intestinal myeloid compartment in either microbial colonized SPF mice, antibiotically pretreated mice, and gnotobiotic mice. Antibiotically treated mice received a triple combination of Amoxicillin, Gentamicin and Cotrimoxazole in the drinking water starting 14 days prior to allo-HCT.

Allo-HCT induced striking changes within the ileal myeloid compartment of SPF mice. Host-type intestinal perivascular (Adamdec1<sup>+</sup>, Apol7<sup>+</sup>, Ecm1<sup>+</sup>) and scavenging (Mertk<sup>+</sup>, Msr1<sup>+</sup>, Wfdc17<sup>+</sup>) tissue-resident macrophages of the lamina propria were maintained, while their submucosal counterpart and dendritic cells vanished within 48 hours after allo-HCT. In contrast, allo-HCT disrupted the ileal myeloid compartment of gnotobiotic mice only transiently, which fully recovered within 6 days. Notably, the intact macrophage network in gnotobiotic mice significantly reduced allogeneic effector T cells in the Ileum compared to SPF mice (3.89-fold decrease, n=6, p=0.0007). Antibiotic pretreatment of SPF animals did not prevent myeloid network disruption as myeloid subsets clustered similarly in both conditions.

Moreover, antibiotic treatment dysregulated macrophage gene expression, triggering PAMP-responsive programs and inflammation indicating a loss of regulatory potential. This became also evident by morphological changes, as macrophages retracted their protrusions and adopted a more rounded shape, as observed in confocal microscopy. Spectral flow cytometry revealed similar changes in other GvHD-target organs such as the colon, liver, and skin, where myeloid cells were significantly reduced or depleted.

Conversely, CD64<sup>+</sup>F4/80<sup>+</sup>PD-L1<sup>hi</sup> kidney macrophage numbers increased after allo-HCT regardless of microbial colonization. Additionally, newly identified CD177<sup>+</sup>PD-L1<sup>+</sup> GMP-derived Ly6C<sup>+</sup> classical monocytes preferentially migrated into the kidney after allo-HCT, rapidly replenishing and increasing PD-L1<sup>hi</sup> tissue resident macrophage numbers. This fostered the macrophage network and likely protected the kidney from an overwhelming aGvHD response.

These results highlight differentially regulated macrophage networks and stabilities in organ specific niches. Targeted expansion and strengthening of tissue resident macrophage networks appear to be an attractive therapeutic approach for mitigating aGvHD in mice and patients.

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